

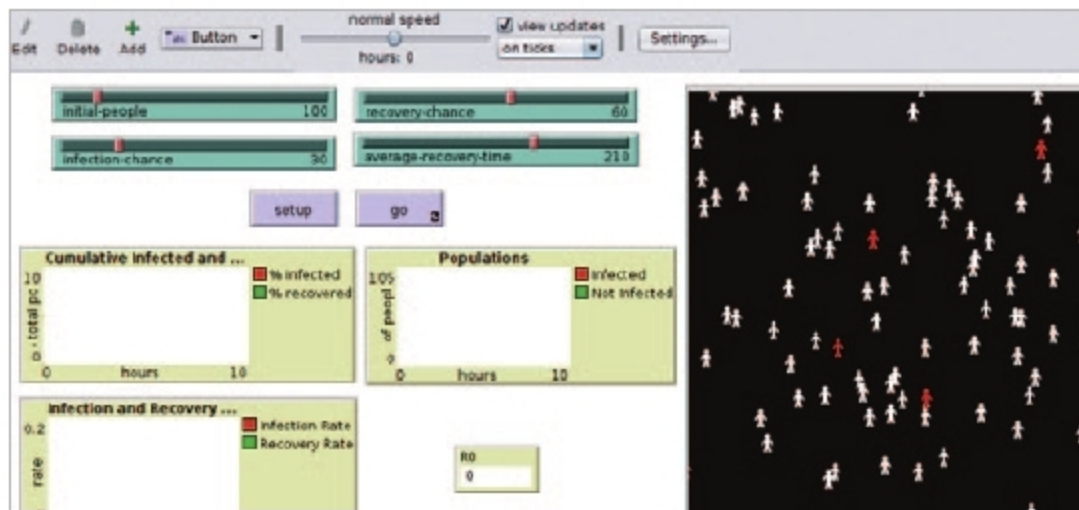


Figure 3: epiDEM – initial situation

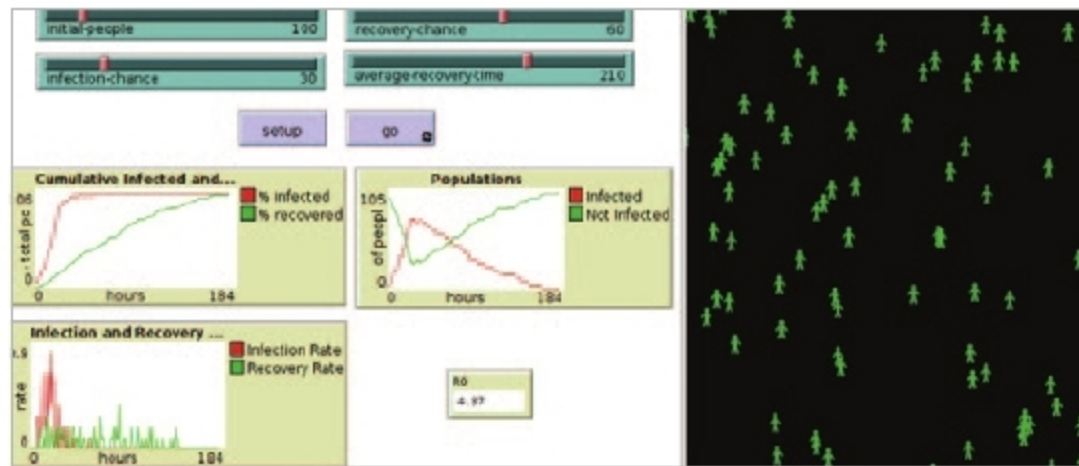


Figure 4: epiDEM – final situation

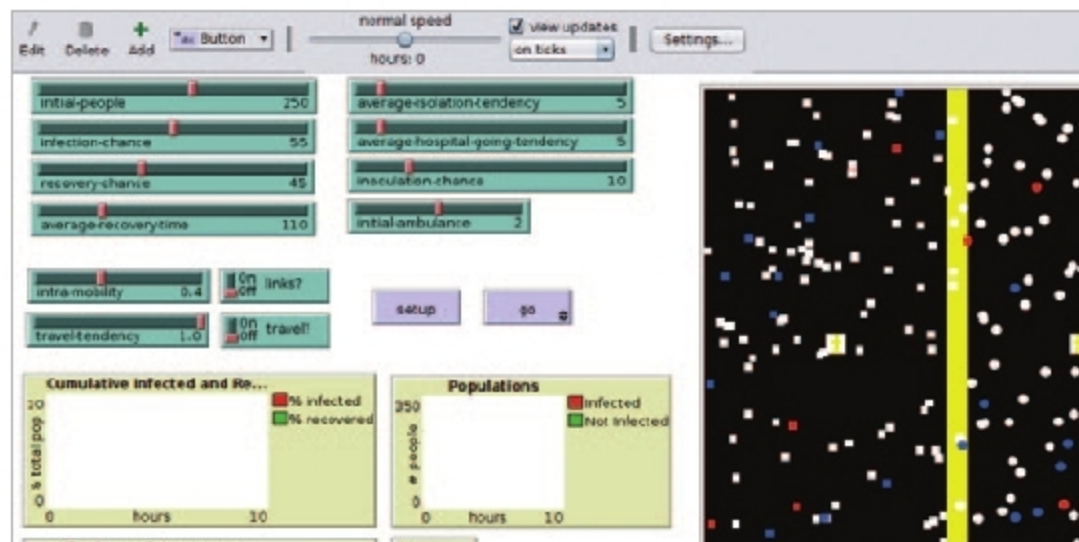


Figure 5: epiDEM Travel and Control – initial situation

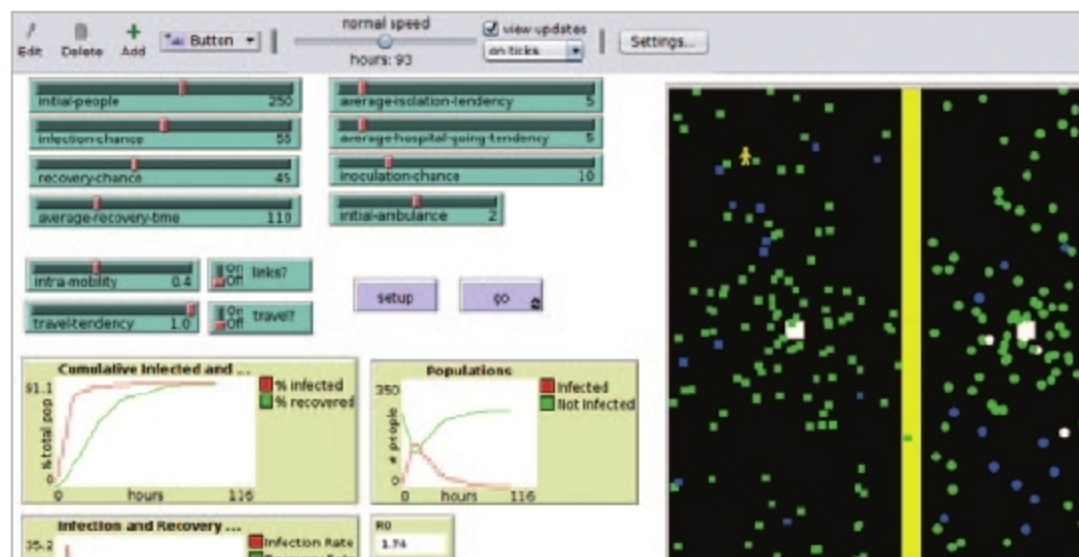


Figure 6: epiDEM Travel and Control – final situation

Disease Solo

Disease Solo can simulate the spread of an outbreak within a population. One agent in population is controlled manually, while others are controlled by the computer. Figure 1 shows the initial situation of a single infected person among a 100 people. After 183 steps, all 101 people are infected, as shown in Figure 2. The graph in Figure 2 looks similar to how the outbreak happened in China.

epiDEM

The Epidemiology Understanding Disease Dynamics and Emergence through Modeling (epiDEM) basically simulates the spread of an infectious disease in a closed population (community/country). Systemic dynamics infection from one person to a susceptible population can be studied. It has an additional feature of gauging the recovery rate. Figure 3 shows how, initially, some people were infected in a population of 100. After 184 steps, Figure 4 shows a recovery rate of 60 per cent.

epiDEM Travel and Control

epiDEM Travel and Control is an extension of the basic model. This models the spread of the disease in a semi-closed population, but with additional features that factor in travel, isolation, quarantine, inoculation, and links between individuals. However, it is assumed that the virus does not mutate and that complete recovery is possible. After 93 hours, Figure 6 shows that people have recovered, compared to Figure 5. It is possible to study the outbreak of a disease within a population in a community or area, with all the available parameters.

There are other models available with various features that include simulating the spread of the disease in a number of different conditions and environments, the interactions of agents, etc. However, we stop here. Netlogo can be installed and simulated with Ubuntu-14.04 LTS OS. **END** 🐧

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